Durham University

# Examination Paper

| Examination Session: | Year: | Exam Code: |
|---|---|---|
| May/June | 2025 | ENGI3341-WE01 |

**Title: (BEng/MEng)**
Environmental Engineering 3

| | |
|---|---|
| Time Allowed: | 2 hours |
| Additional Material provided: | Yes – formula sheet included as part of exam paper. |
| Materials permitted: | N/A |
| Calculators Permitted: | Those from the Casio fx-83 and fx-85 series. |

| **Instructions to Candidates:** | |
|---|---|
| | This is an in-person, invigilated exam.<br><br>Answer ALL questions.<br><br>Use a separate answer book for each question.<br><br>All relevant workings must be shown. |

| Revision: | |
|---|---|

**Q1**

(a) Balance the following dissociation reactions:

i. $CaSO_4(s) \longleftrightarrow$ **[5%]**

ii. $Na_2CO_3(s) \longleftrightarrow$ **[5%]**

iii. $Mg(OH)_2(s) \longleftrightarrow$ **[5%]**

iv. $FeCl_3(s) \longleftrightarrow$ **[5%]**

v. $NH_4Br(s) \longleftrightarrow$ **[5%]**

(b) i. The concentration of $NO_3{}^-$ in a groundwater sample was determined to be 0.0045 moles $L^{-1}$. Convert this concentration to mg $L^{-1}$. **[10%]**

ii. Name the directives which set out the maximum allowable concentration for $NO_3{}^-$ in groundwater and state whether the groundwater sample, described in Question 1bi, is likely to be considered overly contaminated. **[5%]**

(c) i. Explain the ways in which nitrogen can be removed from wastewater during wastewater treatment, providing relevant details concerning biological processes and treatment methods and giving details of the bioreactor configurations which integrate these processes. **[40%]**

ii. Figure Q1 shows how operational parameters impact nitrogen removal in a wastewater treatment facility. Identify how each of the zones, from 1 to 4, impact the activity of nitrifying organisms. **[20%]**

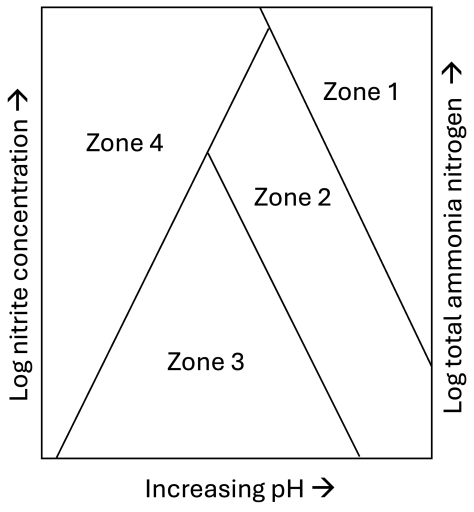



Figure Q1

**Q2**

(a) A large pool of discoloured liquid waste has been found downstream of a large-scale industrial park. The waste was reported to the local authority and an investigator was sent out 3 days later. Upon sampling it was determined that the liquid had a $BOD_5$ of 1260 mg $O_2$ $L^{-1}$ with a rate constant of 0.2 $day^{-1}$. There are three potential sources for the organic waste: Source 1 - a pharmaceutical company (containing organic waste with an ultimate BOD of 560 mg $O_2$ $L^{-1}$); Source 2 - a tannery (containing organic waste with an ultimate BOD of 1990 mg $O_2$ $L^{-1}$); Source 3 - an oil refinery (containing organic waste with an ultimate BOD of 3630 mg $O_2$ $L^{-1}$). Determine the most likely source of the waste. **[35%]**

(b) i. Two decades after the scenario described in Question 2a, the industries have closed, and the site is now considered to be brownfield land (or previously developed land). You have been tasked with assessing if there is a risk that the land is contaminated as per Part 2A of the Environmental Protection Act 1990 (which provides a risk based approach to the identification and remediation of land where contamination poses an unacceptable risk to human health or the environment). Briefly describe how you would assess the situation and what things need to be present to define land as being contaminated and where there is a Significant Possibility of Significant Harm (SPOSH) to human health. **[20%]**

ii. For the example described in Question 2bi, it was determined that the land is contaminated. An options table is outlined below with three remediation strategies. For each strategy provide a simple description of potential advantages or disadvantages as well as an interpretation as to how the remediation option is in line or opposes at least one of the UN Sustainable Development Goals.

| | Containment | Landfill | Bioremediation |
|---|---|---|---|
| Description | A containment barrier is installed to prevent contaminants leaching into nearby waterways | The contaminated soil is excavated and transported to a specialised landfill site | Bacterial and fungal isolates are added to the soil to break down contaminants |
| Time | 9 months | 15 months | 36 months |
| Cost | £250,000 | £750,000 | £400,000 |

**[30%]**

(c) Briefly describe three positive impacts the soil microbiome has on human and ecosystem health. **[15%]**

**Q3**

(a) Water flows at a steady rate, $Q$ [$L^3T^{-1}$], along a horizontal open channel under a sluice gate with an opening of thickness, $a$ [L]. The channel cross-section is rectangular with a breadth, $B$ [L]. Water depths are measured upstream and downstream of the sluice gate and denoted $H_1$ [L] and $H_2$ [L], respectively. Using only equations provided in the accompanying formula sheet and stating assumptions where appropriate, derive an additional equation to predict the open channel flow rate using the two water depth measurements. [**40%**]

(b) Two reservoirs are connected by a cylindrical pipe of 12 cm diameter and 200 m length. The outlet of the pipe is 8 cm below the inlet of the pipe. The Manning coefficient of the pipe is 0.01 $m^{-1/3}s$.

i. Determine the flow rate through the pipe when the pipe is completely filled with water and the water level in the downstream reservoir is 15 cm below the water level in the upstream reservoir. [**30%**]

ii. Determine the flow rate through the pipe when the water depth in the pipe is 4 cm and normal flow conditions apply. [**30%**]

**ENGI3341-WE01 Environmental Engineering 3**
**Formula sheet**

**Polluted Environments**

Ultimate biochemical oxygen demand (BOD):

$$L_0 = \frac{Y_t}{1 - e^{-kt}}$$

Notation:

$L_0$ Ultimate BOD [$ML^{-3}$].

$Y_t$ Amount of BOD removed after time, $t$ [$ML^{-3}$].

$k$ Rate constant [$T^{-1}$].

$t$ Time [T].

The 17 UN Sustainable Development Goals are as follows:

SDG 1 No poverty
SDG 2 Zero hunger
SDG 3 Good health and well-being
SDG 4 Quality education
SDG 5 Gender equality
SDG 6 Clean water and sanitation
SDG 7 Affordable and clean energy
SDG 8 Decent work and economic growth
SDG 9 Industry, innovation and infrastructure
SDG 10 Reduced inequalities
SDG 11 Sustainable cities and communities
SDG 12 Responsible consumption and production
SDG 13 Climate action
SDG 14 Life below water
SDG 15 Life on land
SDG 16 Peace, justice, and strong institutions
SDG 17 Partnerships for the goals

**continued**

Periodic Table of the Elements
1 IA 1A
2 IIA 2A
3 IIIB 3B
4 IVB 4B
5 VB 5B
6 VIB 6B
7 VIIB 7B
8
9 VIII 8
10
11 IB 1B
12 IIB 2B
13 IIIA 3A
14 IVA 4A
15 VA 5A
16 VIA 6A
17 VIIA 7A
18 VIIIA 8A
1 H Hydrogen 1.008
2 He Helium 4.003
3 Li Lithium 6.941
4 Be Beryllium 9.012
5 B Boron 10.811
6 C Carbon 12.011
7 N Nitrogen 14.007
8 O Oxygen 15.999
9 F Fluorine 16.998
10 Ne Neon 20.180
11 Na Sodium 22.990
12 Mg Magnesium 24.305
13 Al Aluminum 26.982
14 Si Silicon 28.086
15 P Phosphorus 30.974
16 S Sulfur 32.066
17 Cl Chlorine 35.453
18 Ar Argon 39.948
19 K Potassium 39.098
20 Ca Calcium 40.078
21 Sc Scandium 44.956
22 Ti Titanium 47.867
23 V Vanadium 50.942
24 Cr Chromium 51.996
25 Mn Manganese 54.938
26 Fe Iron 55.845
27 Co Cobalt 58.933
28 Ni Nickel 58.693
29 Cu Copper 63.546
30 Zn Zinc 65.38
31 Ga Gallium 69.723
32 Ge Germanium 72.631
33 As Arsenic 74.922
34 Se Selenium 78.971
35 Br Bromine 79.904
36 Kr Krypton 83.798
37 Rb Rubidium 85.468
38 Sr Strontium 87.62
39 Y Yttrium 88.906
40 Zr Zirconium 91.224
41 Nb Niobium 92.906
42 Mo Molybdenum 95.95
43 Tc Technetium 98.907
44 Ru Ruthenium 101.07
45 Rh Rhodium 102.906
46 Pd Palladium 106.42
47 Ag Silver 107.868
48 Cd Cadmium 112.414
49 In Indium 114.818
50 Sn Tin 118.711
51 Sb Antimony 121.760
52 Te Tellurium 127.6
53 I Iodine 126.904
54 Xe Xenon 131.294
55 Cs Cesium 132.905
56 Ba Barium 137.328
57-71
72 Hf Hafnium 178.49
73 Ta Tantalum 180.948
74 W Tungsten 183.84
75 Re Rhenium 186.207
76 Os Osmium 190.23
77 Ir Iridium 192.217
78 Pt Platinum 195.085
79 Au Gold 196.967
80 Hg Mercury 200.592
81 Tl Thallium 204.383
82 Pb Lead 207.2
83 Bi Bismuth 208.980
84 Po Polonium [208.982]
85 At Astatine 209.987
86 Rn Radon 222.018
87 Fr Francium 223.020
88 Ra Radium 226.025
89-103
104 Rf Rutherfordium [261]
105 Db Dubnium [262]
106 Sg Seaborgium [266]
107 Bh Bohrium [264]
108 Hs Hassium [269]
109 Mt Meitnerium [278]
110 Ds Darmstadtium [281]
111 Rg Roentgenium [280]
112 Cn Copernicium [285]
113 Nh Nihonium [286]
114 Fl Flerovium [289]
115 Mc Moscovium [289]
116 Lv Livermorium [293]
117 Ts Tennessine [294]
118 Og Oganesson [294]
Lanthanide Series
57 La Lanthanum 138.905
58 Ce Cerium 140.116
59 Pr Praseodymium 140.908
60 Nd Neodymium 144.243
61 Pm Promethium 144.913
62 Sm Samarium 150.36
63 Eu Europium 151.964
64 Gd Gadolinium 157.25
65 Tb Terbium 158.925
66 Dy Dysprosium 162.500
67 Ho Holmium 164.930
68 Er Erbium 167.259
69 Tm Thulium 168.934
70 Yb Ytterbium 173.055
71 Lu Lutetium 174.967
Actinide Series
89 Ac Actinium 227.028
90 Th Thorium 232.038
91 Pa Protactinium 231.036
92 U Uranium 238.029
93 Np Neptunium 237.048
94 Pu Plutonium 244.064
95 Am Americium 243.061
96 Cm Curium 247.070
97 Bk Berkelium 247.070
98 Cf Californium 251.080
99 Es Einsteinium [254]
100 Fm Fermium 257.095
101 Md Mendelevium 258.1
102 No Nobelium 259.101
103 Lr Lawrencium [262]
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**continued**

**Civil Engineering Hydraulics**

The continuity equation:

$$A_1 v_1 = A_2 v_2$$

The momentum equation:

$$F = \rho Q(v_2 - v_1)$$

The Bernoulli equation:

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

Manning's equation:

$$S_f = \frac{n^2 v^2}{m^{4/3}}$$

Partially filled cylindrical pipe geometry:

$$A = \frac{D^2}{4}\left(\theta - \frac{\sin 2\theta}{2}\right)$$

$$P = \theta D$$

$$\theta = \arccos\left(1 - \frac{2H}{D}\right)$$

Notation:

| | |
|---|---|
| $A$ | Cross-sectional area [$L^2$]. |
| $D$ | Diameter [L]. |
| $F$ | Net force in flow direction [$MLT^{-2}$]. |
| $g$ | Gravitational acceleration [$LT^{-2}$]. |
| $H$ | Water depth [L]. |
| $m = \frac{A}{P}$ | Hydraulic radius [L]. |
| $n$ | Manning coefficient [$L^{-1/3}T$]. |
| $p$ | Fluid pressure [$ML^{-1}T^{-2}$]. |
| $P$ | Wetted perimeter [L]. |
| $Q$ | Flow rate [$L^3T^{-1}$]. |
| $S_f$ | Friction slope [-]. |
| $v$ | Fluid velocity [$LT^{-1}$]. |
| $z$ | Elevation [L]. |
| $\rho$ | Fluid density [$ML^{-3}$]. |

**END**